

Digital Twins of Supply Chains: A Systems Approach

Vitor Jesus , Dimitra Kalaitzi , Luciano Batista , Nestor Leal Lopez

Abstract— We tackle the problem of digitalisation of Supply Chains, focusing on collaboration and sharing of information. By generalising the notion of Digital Twins, we review, develop, and conceptualise the (emerging) notion of Digital Twins of Supply Chains (DTofSC). Whereas Digital Twins is an active research area with data available from numerous industry projects, its application to Supply Chains is just emerging as a cross-discipline stemming from decades of maturing the notion of Digital Supply Chains. Whereas Digital Twins has been used in Supply Chains, the notion of digitalisation of Supply Chains is only a nascent area, with imprecise definition beyond key metaphors (e.g., visibility, traceability); as we conceptualise, the overall vision is to create technical and organisational mechanisms enabling any Supply Chain to be monitored and controlled from, e.g., a dashboard screen. After a literature review on the intersections between Digital Supply Chains, Digital Twins, and digital technologies (e.g. Blockchain), we (1) propose a synthesis systems architecture of DTofSC, (2) identify a key requirement gap (that we term addressability), and (3) to ground our contributions and in the absence of real-world use-cases in practice, we apply our conceptualised systems architecture to battery recycling based on feedback from an industry-targeted workshop.

Managerial Relevance— The paper provides valuable insights on how companies can benefit from leveraging digital twin technologies. Digital Twins applications in 'circular' supply chains is particularly considered through an illustrative use-case on how Digital Twins support resource recovery decisions, such as battery recycling. Applied to the management of supply chains, Digital Twins provide an amalgamation of digital replica and data that can help managers gain insights on the behaviour of their supply chains and make managerial decisions to mitigate supply disruptions and operational discrepancies against critical performance analytics such as lead times, inventory levels, demand fluctuations, sustainability impacts, and other key performance indicators. Digital Twins relevance for supply chain risk management is also considered. By using risk profiles based on real-time data analytics combined with historical performance of vendors, materials, routes, macro environment, and so forth, managers can perform simulations with modified parameters such as increased rainfall, import sanctions, infrastructural obstructions, etc., and make decisions to improve the resilience of their supply chains by anticipating likely disruptions. The range of interrelated variables to be considered in such complex environment presents an opportunity to apply AI techniques to improve Digital Twins models supporting supply chain management decisions.

Index Terms— Supply Chains, Digital Twins, Digital Supply Chains, Digital Twins of Supply Chains, Blockchains

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I. INTRODUCTION

DIGITALISATION is now an established context across society and the economy, with the progressive integration of Information and Internet-based technologies since the late 1990s. Nevertheless, one area that, persistently resists comprehensive digitalisation is Supply Chain Management (SCM). Whereas *internally* to a firm we see wide usage of technology and communications, when multiple parties need to *collaborate* – the essence of a Supply Chain (SC) [1] – we still see an overreliance on manual processes [2] such as finding suppliers, tracing materials and parts, or coordinating international logistics.

Consider the case of Food Manufacturing: still today, it is practically impossible for a consumer to significantly trace ingredients of a ready-made meal, despite consumer pressure and regulations. Another example, later discussed in this paper, is that of recycling used batteries, a pressing problem for sustainability reasons but also due to limited availability of raw materials. Nearly always, participants in a SC have limited visibility [3] beyond its direct suppliers or customers. Notable exceptions exist, with parcel delivery for end consumers at the forefront where a package may be tracked down to a few streets but even in this case systems mostly work efficiently within a single logistics operator. This state of things is problematic for several reasons, with efficiency and robustness at the forefront, but also due to its direct impact on sustainability and the much-needed aim to make supply chains increasingly circular.

The notion of Digital Supply Chain is still imprecise [4] but we define it as *the ability to use digital technologies to observe and actuate on a Supply Chain*, both in real-time and to support risk management. Leading companies such as Amazon, IBM, and Microsoft have heavily invested in the implementation of Digital twins as they can offer numerous advantages in various applications, including predicting product failures, raw material usage and waste analysis, real-time feedback provision, etc. [5]. For example, DHL has created a digital replica of Tetra Pak's warehouse in Singapore as part of its supply chain management system in Asia Pacific and Mars Wrigley and Accenture implemented a Digital Twin to simulate the movement of chocolate and other materials during the manufacturing process which enabled them to pinpoint areas of congestion and inefficiency, leading to a more efficient production process [6][7]. However, we differentiate from existing work on that we

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focus on the Digital Twins of the Supply Chain rather than use of Digital Twins in Supply Chains. In other words, we model the processes of a Supply Chain and not how Supply Chains can be made more efficient by using Digital Twins of the physical products or systems. We stress that our approach envisions well beyond limited integrations of IT systems between suppliers and customers – e.g., email or web procurement platforms. We here include the ability to visualise and control an existing or potential (e.g., not yet formed) supply chain.

Our approach is to look at Supply Chains from a Computer Networks/Services perspective; in this sense, we adopt a model of a SC that is a many-party, self-organising, decentralised (no central coordination points) network of physical and digital entities. By entities, we mean goods, materials, parts, components, and packaged products, but also information and monetary flows, human work, associated documentation, events, etc. Similar to the Internet-of-Things, where a device (such as a home thermostat or a robot remotely controlled) generates and receives data, we explore the notion that a SC and its entities can have a mirror representation that, depending on its nature, is both readable and writable – i.e., receives data and accepts requests to change. To this end, a central concept we see now developing and expanding beyond product manufacturing is *Digital Twins* (DT). Similarly to the notion of Digital Supply Chain, its precise definition is still to be established [7], particularly outside product design or manufacturing. In its simplest form, DTs currently are mostly associated to the idea of a physical object having a virtual representation – e.g., a physical robot operating on a manufacturing line that can be remotely observed and controlled. DTs can be extended to non-physical processes, including those depending on human interaction, and outside a single organisation, which raises the notion of *Digital Twins of Supply Chains* (DTofSC), the central concept in this paper.

Whereas this concept is not strictly new, as later reviewed, this is a nascent area needing research and systematisation. The recent literature currently sits at early conceptual steps (e.g., [1][2][3][4]), with scarce use-cases focusing on Digital Twins of the products and/or small steps in production, while the Supply Chain processes in themselves are not fully considered. Similar to digitally observing and interacting with a robot in a manufacturing line, we here envision similarly featured SCs: operators have access to the whole SC in near-real-time from, e.g., a screen and a connection to the Internet. Our central research question is thus: *To which extent can we generalise the notion of Digital Twins in product design/manufacturing to Supply Chain Management?*

We note that, rather obviously, there is a fundamental difference between trying to model a physical, self-contained process, and trying to model a supply chain. As alluded to, SC automation can only reach a certain realistic level and many steps are human-based. For example, on delivery of a packaged battery from a distant supplier, physical transportation and some form of signature will be required. Once the battery is integrated in the destination system, it is very difficult to trace it once disassembled, a well-known problem in battery recycling [7][9], which is not rooted in technology but, we

argue, in the lack of widely agreed system models and implementations. As we will elaborate, a significant barrier that has not been raised before, to the best of our knowledge, is that entities of a generic supply chain (parts, materials, transportation cars, international borders' documentation, etc.) are required to be computer-addressable, similar to an internet server. The underlying metaphor is that, e.g., a battery should have its own unique and persistent address in a global network, which is the SC. This address will enable (with the appropriate technologies and system architectures), at any moment or location, the object to be queried and actuated on, similar to accessing a shared document on the Internet. Since, naturally, it is not expected that parts or goods, or even human processes, to have any kind of connectivity in the technical sense, a human will need to “check-in” at key events. Overall, the interface between human and the generation and management of unique, global identifiers for parts, good, materials, processes, etc., is a key requirement in a DTofSC, and that we will call the addressability requirement. We argue that the combination of DTs, DTofSCs, and the notion of SC addressability can greatly facilitate long desired requirements such as establishing the provenance of products or real-time action on disruptions.

A. Contributions and Methodology

Motivated by accelerated maturing of the DT concept, but realising the notion of DTofSCs is only emerging, our paper offers a combination of two aspects. First, we review the (sparse) literature on approaches to DTofSCs, from system conceptualisation to envisioned applications and benefits. Secondly, we attempt both a generalisation of DTs for SCs, by looking at past results, and a synthesis to develop a generic architecture to DTofSCs. During this exercise, we will note that the intrinsic differences between a SC and a classic example of DTs, requires looking at SCs as a distributed, decentralised network of many parties – that is, there is no coordination point expected. In particular, we identify a specific requirement – addressability – loosely corresponding to the idea that any entity in a SC needs to have its own identity. Combining the extracted insight from literature, we then propose a high-level system architecture. This paper has thus the following overarching goals:

1. Review and generalise the notion of Digital Twins to apply to Supply Chains, aligned with a review of enabling technologies;
2. Identify, position and characterise gaps, particularly, what we call the addressability;
3. Synthesise and propose a generic architecture.

Whereas we attempt a generic conceptualisation of DTofSCs, thus applicable to any SC, we will apply our views the case of battery recycling. This is for two reasons. First, this application has high societal relevance; it is one of the best cases where circularity is being actively researched thus creating a more generic and representative SC. Second, a key challenge is in having access to the specific information of the product at the point of recycling – where the concept of Digital Product Passports (DPP) is being explored, as we will discuss.

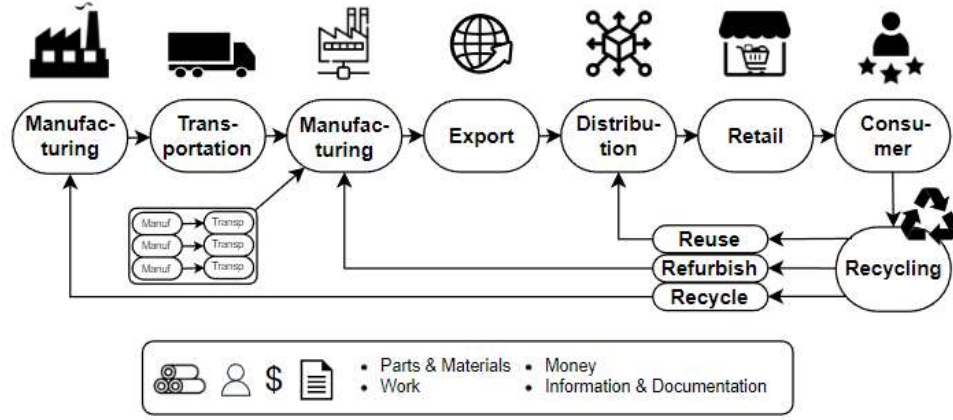


Figure 1. A generic illustration of a (circular) Supply Chain.

We straightforwardly link DPPs to our addressability requirement since it can be the vehicle by which a unique identity of the particular item (thus becoming addressable) is conveyed: the physical battery item entering recycling.

We utilise a mixed methodology. We start by comprehensively reviewing and integrating literature across several key areas: (1) digital supply chains and enabling technologies, (2) digital twins, (3) applications of Digital Twins to Supply Chain management, with a particular focus on blockchains given their prominence. We then perform a synthesis exercise of DTs architectures to propose a generic architecture for DTofSCs. Considering our use-case of battery recycling, a key example in Circular Supply Chains, we reviewed approaches for this particular case, including the mapping and translation of Product Passports into our approach.

The remainder of our paper is as follows. In Section II we review the concept of Digital Supply Chains, along with that of Digital Twins. We then review technologies in Section III supporting or enabling the notion of DTofSCs with a special focus on blockchain technology. In Section IV we synthesise our critical review of literature to propose an architecture of DTofSCs and identify the addressability requirement. We further discuss how to apply our proposal to battery recycling. Section V will layout a research agenda and draw our final conclusions.

II. DIGITAL TWINS OF SUPPLY CHAINS

In this section we review Supply Chains taking an angle of computer systems/networks by drawing parallels with distributed and decentralised systems, to which the Internet as we know it belongs to. A. Supply Chain Management Supply Chain Management is now a well-established and intensively studied concept. It comprises the networks and flows by which products, materials, money, and information flow between manufacturers and end consumers. The notion and architectures of SC have naturally changed over time [4], evolving from a procurement-centred perspective to the current coordination of many parties, each delivering value in a processing chain, often long and geographically distributed. Whereas modern SCs exhibit high degrees of agility and

resilience, this in itself has raised complexities as exemplified by the ripple effect [10] where a delay at any point on a supply chain has repercussions (e.g., inventories) both downstream (towards the end customers) and upstream (towards part manufacturers).

Figure 1 illustrates a generic supply chain, from the perspective of roles. One can imagine a Li-ion battery lifecycle from raw materials to reaching the final customer (a purchased mobile phone or an electric vehicle (EV) sold at a car dealer), to re-entering the supply chain via recycling, refurbishing or reuse [11]. A first manufacturer supplies raw materials (such as Lithium) directly from (e.g.) mining the subsoil. The materials are transported to another manufacturer who, by combining raw materials and parts from other suppliers (e.g., copper cathodes, plastic casing, etc.) creates the final product. This is represented in Figure 1 on the left-middle with a stack of other manufacturers and transportation. The final product could then be exported, distributed to different locations until it reaches retailers who can sell it directly to the end consumer. Note that many different variations exist; the manufacturer of an EV, for example, is itself another manufacturer who is both the end consumer of the previous suppliers and the supplier to end users or other businesses. Such connectivity and interdependence illustrate the complexity of modern Supply Chains.

We should further note, as illustrated at the bottom of Figure 1, that all stages will involve, to varying degrees, four (generic) inputs and outputs: (i) parts, materials and goods, (ii) work, either human or with automation, such as assembling the final Li-ion battery, (iii) money and financial flows, and (iv) information and documentation, often the root cause of inefficiencies and disturbances. The notion of *circular supply chains* is relatively recent [12] but crucial to meet sustainability goals. It consists of inserting a feedback loop from consumers to manufacturers. Circular Supply Chains (CSC) structure the way disposed products can be reutilised, repurposed, refurbished, or recycled. Circularity is always driven by sustainability considerations, however, in the case of Li-ion batteries, it is a further necessity [13] given forecasts for the substitution of internal combustion engines with electrical engines.

B. Supply Chains as Many-Party, Decentralised Systems

Taking now a computer-systems perspective, we see that a SC can be modelled as a mathematical graph (a “network”) with multiple *nodes* (the participants) and *edges* (connections between nodes), with each edge receiving a multi-dimensional weight or profile. Over this graph, an all-seeing optimisation function – often known, in computing terms, as an *oracle* – would be able to determine, at any moment, the best connectivity path (e.g., for a desired product profile, which parties to engage and by which order) so that the required output is feasible and all participants locally (in the absence of high level policies) maximise their inputs (e.g., financial rewards) considering a global equilibrium point. Beyond optimisation, visualisations and real-time control are possible. At this level of simplification, this is a problem that computer networks have solved and operationalised decades ago – e.g., servers were manually added in the early days of the internet by maintain a list of people who would telephone each other to notify of any changes.

However, for such an all-seeing optimisation oracle to exist, a number of conditions need to be met which are often unfeasible when looking at SCs and compared to network systems. For example, information does not flow quickly and often there is limited visibility beyond the immediate SC echelon. Furthermore, discovery of all participants (unlike, e.g., Internet routers) is not trivial and is a challenge on its own [14]; events leading to disruption are also difficult to communicate and broadcast given the common levels of automation.

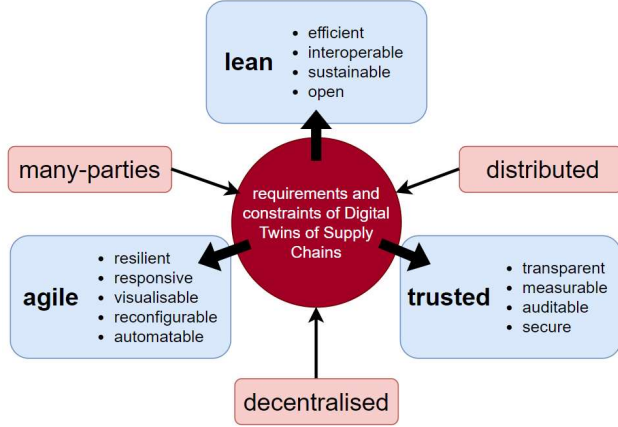


Figure 2. Requirements of Digital Supply Chains – Computer networks vs Operations management perspectives.

Following this perspective, the key challenge of modern SCs could be formulated as discovering and coordinating a distributed, decentralised, many-party network:

- *Distributed*, as the graph can change at any point and participants (within a role) are mutually replaceable (to varying extents)
- *Decentralised*, as there is no global entity coordinating
- *Many-party and open*, as participants are many and often enter and leave the graph randomly.

From an Operations Management perspective, but as drawn from the angle of Digital Supply Chains, the following key

requirements should be met and/or promoted by digital technology:

- *lean* [14][15], in the sense of being efficient, interoperable, sustainable, and open [7][9][16][17][18][1][19][20][21][22];
- *agile* [14][17][1], in the sense of being resilient, responsive, visualisable, reconfigurable, automatable, and allowing (near) real-time communication [23][16][18][1][19][20][23][1];
- *trusted*, which encompasses requirements such as being transparent [16][24], measurable (real-time, short, and long term trends) and auditable [1][16][19][20][23][26][27], and secure against malicious or opportunistic actors [16][24][25].

C. Integrating Computer vs Operations Perspectives

The identified requirements from Operations Management disciplines should now be confronted with properties as derived from computer networks – Figure 2. Whereas the requirements of a Digital Supply Chain derive from the physical properties of Supply Chains, they raise complexities given key constraints: a large number of potential parties participating in a specific SC, over a network that is intrinsically distributed and is necessary decentralised. We argue that it is the set of these constraints that makes blockchain technologies so immediately appealing. Blockchains are natively decentralised, distributed and, in principle, able to cope with many parties as anyone can use it as a central point acting as a coordinating point. However, as we will argue next, blockchain technologies do not necessarily meet all requirements.

III. TECHNOLOGIES FOR DIGITAL SUPPLY CHAINS

Literature is currently scarce on the notion of Digital Twins for Supply Chains. The specific topic of Digital Supply Chains has been explored for more than a decade, but it is often narrow-scoped – e.g., digital collaboration between adjacent SC tiers or customers [26], the (local) use of Enterprise Resource Planning Systems (ERP) [15], or specific to an industry or sector (e.g., automotive [28]). On one hand, it directly ties to the development of the underlying concept of Digital Twins, and, on the other hand, needs specific re-conceptualisations, generalisations, and adaptations.

We review three key technical dimensions. The first is the utilisation of blockchains, prominent in literature in the last decade, along with integrations with other technologies. The second is the specific contexts of Digital Supply Chains and Digital Twins, and its intersections. We finally review a particular aspect that we argue is crucial for the development of DTofSCs, data models, as it will become apparent when we propose our architecture of DTofSCs.

A. Blockchain Technology

Since the concept of Blockchain was introduced with Bitcoin in 2009, Supply Chain management saw in it great potential to solve long standing problems, particularly about visibility and

provenance. About a decade past, the researcher and practitioner communities learned valuable lessons and now the technology and its utilisation in SCM is clearer and more mature, albeit still with important questions needing research up to clarifying misconceptions about its practical applicability [29]. Historically, we see marked increasing research activity from 2017 onwards [30][71], as judged by publication volume and new journals.

Whereas there is significant technical complexity in blockchain-based applications and services, the overall concept is simple. Figure 3 compares (simplified) models for SC collaboration, namely, (1) generic, Internet-based, Electronic Data Exchange (EDI) technologies (often, a common website and database) and (2) a blockchain. From this perspective, a (public) blockchain is a global, public, single copy of a digital ledger which only accepts appending of data. Further, by using cryptography and a vast number of anonymous parties distributedly operating the blockchain (e.g., Bitcoin miners or Ethereum 2.0 validators), it is practically impossible, after a relatively short period of time (hours) to modify the appended information. This property is sometimes called irreversibility or append-only [31]. Once a new record is pushed into the blockchain, it becomes visible to anyone within (typically) minutes.

It thus, acts, as a *single point of contact*, with global visibility, and *avoids service/functionality discovery*, a common complexity that is not always brought forward in system design. It is the combination of these two properties that allows its use as a digital currency, as is the case of Bitcoin, where a simple 2-column table (the ledger of addresses versus amount) is kept, elegantly solving the double-spend problem for which no realistic solution existed before Bitcoin. A transaction is thus a mere update to this ledger. In the case of Ethereum, the information stored goes beyond a simple 2-column ledger and allows logic (e.g., mathematical operations or test of conditions) to be implemented. In general, both the code of the specific application, and the state (e.g., program variables) are publicly auditable enabling seamless collaboration and transparency.

Therefore, its trust model is strong, especially noting that there are no intermediaries or “technical administrators” who need to be part of the trust model. In contrast, an EDI-based model requires each party to discover others ahead of collaboration, to find interoperable interfacing points. Nearly always, this requires a mediator party to develop code and functionality (e.g., a collaboration platform) to handle collaboration data – and, equally nearly always, this acts as a black box and therefore not auditable. Its security model is therefore much weaker, particularly in terms of information integrity, as it is directly dependent on the (single or few) parties operating the interoperability platform.

These intrinsic properties of blockchain technologies – (i) global visibility and audibility, (ii) single point of interaction, and (iii) cryptographically immutable appending of data – make (public) blockchain a unique technology when it comes to collaboration formats such as trustless operations, transparency,

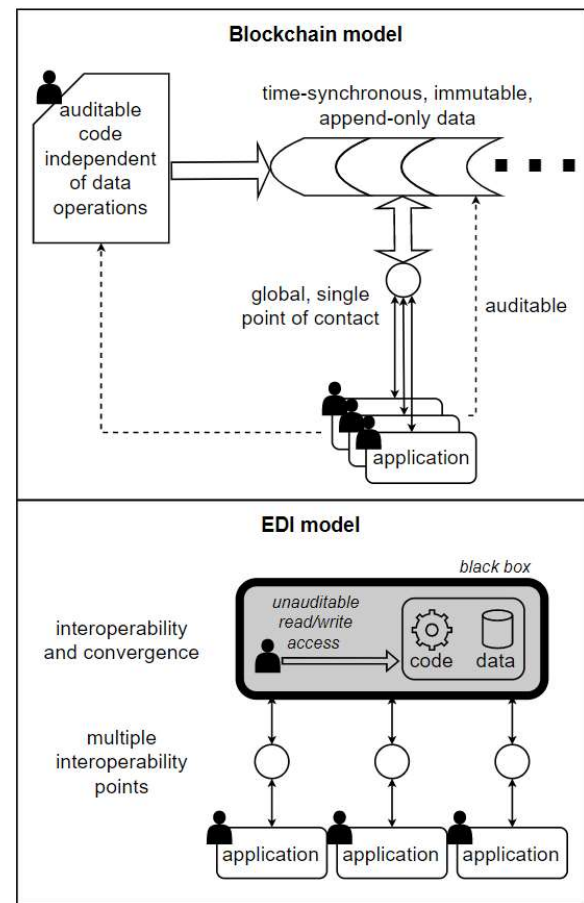


Figure 3. Comparison between Blockchain and EDI models for Supply Chain collaboration.

disintermediation, or provenance, traceability and trackability [32]. We note that permissioned blockchains essentially follow the model of EDI (and thus inheriting its weaker security model), which may explain why its popularity is seemingly in decline for new projects when compared to implementations based on public blockchains.

Whereas the three mentioned properties of blockchains have a definite role in Digital Supply Chains, we argue that they must be considered a component that integrates with other technologies in a wider model. In particular, its underlying trust model is not generally extendable to the wider system [25]. For example, whether one can trust that the data in the ledger has not been modified and thus represents a “ground truth”, one cannot make assumption about, e.g., the quality and trustworthiness of the data itself.

From a digital supply chain perspective, we can break down literature on the utilisation of blockchain technology in the following key areas: (i) use-cases, applicability and potential, (ii) technology integrations and systems, and (iii) the human and managerial element, including perceptions and product development complexity.

1) Use-Cases, Applicability and Potential

There is now a significant volume of literature on use-cases and literature reviews (e.g., [29][32][33][14][34][35][36][37]).

Although challenges exist [33][38][39], virtually all areas report benefits in integrating blockchains. Table 1 maps literature to application areas with an outline of the envisioned or potential impact as reported. In terms of applicability, one can broadly identify three themes in literature: (i) online collaboration, (ii) trusted information, (iii) disintermediation.

Online Collaboration— An important theme reported in literature, that has transversal importance across virtually all sectors, is Provenance, Tracking and Tracing: whereas provenance relates to establishing the origin and authenticity of products (further discussed in the next theme), tracking enables (near-) real-time updates of the journey of goods and tracing the ability to see its past handling [32][40][42][44][45][46]. We see transversal operational synchronisation across parties [48] (as goods traverses the SC) as a key feature enabled by blockchain technology. Beyond the ledger, some blockchains (notably Ethereum) support automations [47][48][49]. Whereas Bitcoin is only able to handle an update to the ledger and storing limited information (technically, up to 40 bytes in the original specification of Bitcoin) which can still be used for automation outside the blockchain [51], Ethereum operates around executable code (smart-contracts) making it far more flexible and powerful.

For example, Westerkamp et al [50] propose that meal manufacturers create a digital representation in the blockchain for each product, which is itself composed of ingredients also with individual representation in the blockchain; thus, a meal is a combination of objects and a set of smart-contracts in the blockchain that accept software state changes (e.g., a specific meal or ingredient, which is a smart-contract on the blockchain, is associated with a particular location). In the case of battery manufacturing, if each component/supplier has a unique representation in the blockchain, tracking and tracing becomes trivial, including great visibility (up to in real-time) of bottlenecks or inventory problems. Furthermore, safety recalls can be issued by messaging the smart-contract which can notify owners or retailers. Finally, we also see how the blockchain, being a globally, publicly accessible location (an address of a smart-contract might be sufficient), coordinating multiple parties (suppliers, retailers, end customers, etc.) becomes much easier from a technical standpoint.

These examples illustrate the level of automation that smart-contracts enable while potentially storing (or linking to) equally rich data analytics that can be used in future decision-making. Furthermore, because of the intrinsic association of blockchains and digital money (in the form of cryptocurrencies), payments can be straightforwardly automated [34] reducing indirect costs and effort.

Trusted Information— The unique property of data immutability, past trust in the updating process and sufficient time elapses, enables the retrieval of information that can be trusted in absolute in terms of having not been tampered with. Furthermore, secure and granular access, which is publicly auditable, is an intrinsic and native feature in any public blockchain: in order to modify data at a particular address in the

Table 1. Envisioned benefits of Blockchains per area.

Logistics/Transportation [39][43][44][33][34][35][68][71][72] • Transversal visibility of journeys and storage • Increased synchronisation across transport and logistics • Tracking of controlled characteristics (e.g., temperature) • Facilitated financial flows (e.g., cryptocurrency) • Dematerialisation of documentation • Provision of proof of delivery • Automation of trade (processes, invoicing, etc.) • Inventory management and traceability
Food, Agriculture and Fishing [39][34][68][80] • Trust management • Records of provenance and traceability “farm to fork” • Safety and compliance support • Ingredient traceability and informed consumers • Traceability of “cold chains” • Forward/reverse communication of safety recalls
Manufacturing [39][34][68][33][52] • Inventory management • Goods/parts tracking and shipment • Visibility of carbon footprint • Risk management and mitigation • Anti-tampering, counterfeiting, certification • Support to novel (re)manufacturing approaches such as Additive Manufacturing or Social Manufacturing
Pharmaceuticals [39][34][68][33][52] • Transparency and visibility of manufacturer • Data security, authenticity of clinical trials and patient confidentiality • Quick response and emergency action • Counterfeiting and anti-tampering • Dematerialisation of compliance and regulatory processes
Construction, Built Environment [39][34][68][33][52] • Near-real-time management of resources and suppliers • Visibility of carbon footprint • Project management and party scheduling • Material certification and provenance • Management of title ownership • Trust, liability, auditability • Coordination of a highly fragmented sector • Fast payments support linked to closure of tasks
Others [39][68][43][33][61][62][34][64][73][76] • Proof of ownership and/or possession • Corrections of information asymmetry • Disintermediation • Support to Supply chain finance via trackability • Lower entry barriers via decentralisation • Collaborative manufacturing

blockchain, one needs to have secure cryptographic keys, a strong form of access control (so that the originator of the data can give selective access to selected parties). The data in itself

is not necessarily trusted but, if a party had access to a specific resource on the blockchain and executed an update (such as a smart contract), any other party can safely assume authenticity of data – the first-hand, “ground” truth. An example is cold chains in the Food sector requiring controlled transportation environments, which is tightly regulated [52], or the pharmaceutical sector [53]. Therefore, demonstration of compliance across the supply chain can be more easily supported.

Since the original data is public and ordered in time and by handling party, and can be trusted, data analytics techniques can be applied across the supply chain. A striking example is calculating the carbon footprint of a product, or simply understanding its origins [54], as long sufficient checkpoints exist on the blockchain. The same data can be used to better forecast risk raising the resilience of the particular supply chain and parties [40].

Overall, blockchains can support the underlying goals of sustainability (even if the technology itself may raise challenges [7][9][53]) by facilitating the reconfiguration of supply chains towards circularity [57][58], across a number of critical success factors such as visibility, efficiency and accountability [7][9][59][61][79]. Counterfeiting deterrence and product certification (further combined with provenance assertion) is another envisioned application stemming from the unique properties of blockchains. To this end, blockchains can greatly support processes requiring proof of ownership/possession (e.g., real-estate property titles or art authenticity certificates [76] by leveraging trusted information.

Disintermediation— The third theme in literature concerns the ability to remove third-parties from a supply chain, either coordinating agents (thus making it more decentralised [68][69][70]) or intermediary ones (making it more efficient and short [16][46]). Referring to Figure 3, this directly stems from the fact that a blockchain can be modelled as having a globally unique access point where different parties can virtually meet to collaborate. For example, a small fishing business has now the ability to more easily find and collaborate with a food retailer directly without needing a discovery and mediating party, a source of multiple costs. Contributing to this, and from an economics perspective, blockchain technology is seemingly able to reduce the information asymmetry across the supply chain [68] thus decreasing inefficiencies that potential benefit whole sectors [68][77]. Finally, we also note the potential in social manufacturing and customisation [17] (up to directly meeting end customers), directly linked to disintermediation. Similarly, the sharing economy could benefit [62]. We here include in the theme of disintermediation the ability to dematerialise, that is, remove local or cross-border records, often on paper. For example, one of the first large scale projects recorded was the use of blockchains for cross-border maritime logistics documentation [40].

2) Associated Technologies or Integrations

Blockchain technology can only meet certain requirements if integrated in a larger system for SCs so other technologies are

often reported in association, either in a supplementary or complementary role. Two themes emerge in literature.

The first is its association with sourcing the original data to store on the blockchain. Referring to Figure 3, a key challenge consists of uploading information in the blockchain, a difficult security problem that can compromise the authenticity of on-chain data. Since our context is on goods, that most often are simply packages of raw materials/parts, the challenge is to report on the right object, ideally in an automated fashion. This nearly always will require some level of human intervention even if to confirm the data recorded pertains to the correct physical object. Two technologies thus play a central complementary role. The first is the Internet-of-Things [52][73], which, particularly in scenarios involving automation (e.g., a factory or logistics facility) [78], comprising networked devices (programmable services, sensors and actuators), can be decisive if blockchain technology is to have widespread usage and support its expected merits [43][49]. The second technology concerns tagging of goods. Common technologies use either Quick Response (QR) or bar codes which are a simple (hence cost effective, often on paper) passive technique [38][44][49][59], or the more complex RFID tags, which have electronics that, on receiving a signal, respond with a unique (within the local scope) code or data [16][32][40][43][80].

The second key complementary technology concerns the integration of blockchains with existing digital/IT technology already in existence which we here broadly call ERPs (Enterprise Resource Planning tools). It commonly supports resource planning [81] and supplier coordination [16]. In other words, blockchains are, in themselves, an invisible technology [81] that, e.g., a manufacturing planner operator will not directly interface. Therefore, the IT systems in use need to have an interface so that the blockchain becomes an integral component of the wider IT infrastructure. Furthermore, considering the enhancement in collaboration and visibility potentially offered by blockchain technology, the scope of ERPs needs to be expanded [68][42].

3) Human and Managerial Element

Blockchain technology, and particularly its applications to Supply Chain Management is undergoing a maturity phase where challenges and applications have been better identified, often met with scepticism, particularly at the practitioner’s and managerial level [29]. Beyond the technical suitability and business benefit, there are several challenges involved stemming from the management side [55][57]. On one hand, several research directions, and data from practice, are needed for a truly informed application such as integration with current management theories [16], understanding of incentive/barriers interplay [37][70][82] (particularly in cases where the technology was abandoned [40]), or its associated total cost of ownership and operation and migration [33], including its secondary risks [53][73] when assuming its direct benefits such as trust, speed of collaboration or disintermediation – in other words, whether the technology meets expectations in real-world projects and over time and across different cultures or whether conventional technologies can meet requirements.

Beyond the core technology, the associated effort in technical development should be contemplated, from people to ERP integrations, to the effort in coordinating all parties into adopting blockchain technology. Whereas the effort in coordination, as previously discussed, can be significantly reduced, there is still a significant change to which practitioners may not be open, motivating, e.g., the design of incentive systems [37][74][75]. The software development lifecycle, in itself, is perhaps the biggest barrier for which research is needed. When compared to standard software development (e.g., enterprise applications, mobile and web), software development for blockchains is a new area for which there is a skills gap (since it's a new area), particularly severe when it comes to security, a paradoxical thought at first sight considering that security is strongly associated with blockchains. As repeatedly found in cryptocurrencies and decentralised finance, with multiple high profile and high volume cybersecurity attacks, developing applications is far from straightforward, even if the appropriate skills and human resources can be found, and the right regulatory framework exists [77]. Furthermore, whereas trust also stems from transparency of code, each organisation must still have the appropriate ability to audit code [62], which may not be trivial for small organisations.

4) Critique

Whereas blockchain technology is unique at solving specific technical problems no other technology does, one must be mindful that it is only a technical building block whose features need to be mapped on clearly identified requirements. Furthermore, the overall context of where/how the technology is to be used plays an equally important role. For example, the security of blockchains depend on a much wider project context and its intrinsic irreversibility features should not be extrapolated beyond the specific remit of the system and its requirements. To this end, equivalent technologies – such as the cloud – might be a better alternative.

We argue there are significant misunderstandings about blockchains, particularly illustrated with the phenomenon of permissioned blockchains which are now seemingly in disuse. Other utilisations need to be carefully considered. In the classical application to provenance challenges, if the key requirement is availability of data that is somehow endorsed by (necessarily) a single party (the original manufacturer or seller), a central server in the cloud might be a much more practical solution since the trust model is weak at nearly all stages (tagging the object, recording the journey, etc.). For example, trusted tagging of objects is a technical challenge as tags in themselves can be modified, reused, or replaced. In general, the argument that blockchains create trustless systems needs scope and more clarity and experimentation [62].

B. Digital Twins and Digital Twins of Supply Chains

Whereas the notion of Digital Twin (DT) has been discussed for more than a decade and is now maturing both in theory and in practice, it is mostly seen in manufacturing settings to create

a digital view of a physical product. A number of literature reviews on the strict concept of DTs exist [83][84] [85][86] but the generalisation of the concept to both intangible assets (such as a processes) and cross-organisation contexts is a nascent area for which literature is sparse.

Digital Twins Concept— The notion of Digital Twin is historically associated to having a “virtual representation” of a physical product [83]. Since then, particularly after 2015 [83], the concept has grown in interest in both industry and research while expanding on its scope, up to existing standardisation initiatives such as ISO/IEC 30173 (work in progress) [87]. Naturally, the original notion was focused on simulation and data (from sensors and actuators) [84] leading to either optimisation or visualisation in real-time. A simple model adapted from Tao et al [85] that we use as a baseline for our own, is represented in Figure 4. A physical entity has a digital representation (the Virtual Entity or Digital Twin) that represents, with varying fidelity, the physical object, potentially being able to control and actuate on the physical object. Data collected is stored and fed into higher-level services such as remote visualisation or towards optimisation processes. The arrows represent flows of communication such as data over a network or the Internet.

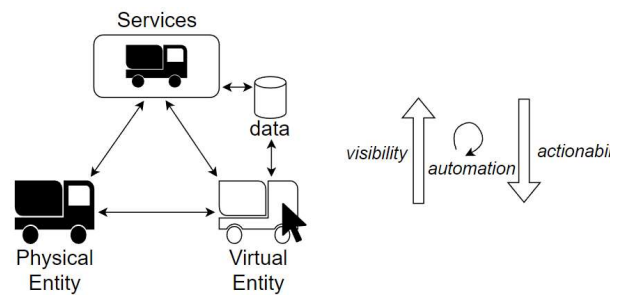


Figure 4. Simplified DT model.

DTs are in themselves a product of new technologies such as Virtual Reality, Big Data or IoT [88][85] enabling new forms of managing products which, in turn, raise a number of technical challenges. From the simpler perspective of a physical/virtual modelling, we see challenges around fidelity between the physical and digital twins, synchronisation of state and data, reconfigurability and model reuse across different products or systems, calibration and reliability of data and control, etc. [89]. From a systems perspective (e.g., a farm or a smart city), the complexity significantly increases, as we see challenges including modelling techniques (physical [83] and in terms of visual representation and human interactivity [84]), leveraging existing simulation and optimisation strategies [85], data fusion and related granularity, synchronisation rates) [85][84], adaptation to networked control [85][84], data models and data management [84], interoperability [84] (including twin-to-twin [85]), cybersecurity [86], etc.

A generalised conceptualisation is not trivial, given that DTs across different application areas show different challenges, some even specific to the use-case – e.g., compare

modelling an airplane engine to Vertical Farming. As such, one could argue that the area is being driven by successive cycles of experimentation (including commercial projects) and synthesis [90][91]. For example, Farming is seeing wide adoption where environmental conditions can be tightly supervised [92]; another example, for similar reasons, is energy systems [94].

Digital Twins of Supply Chains—Beyond modelling physical settings, a recent strand of research concerns applying Digital Twins to intangible settings (or a mix) such as logistics, circularity or whole supply chains. Whereas the term Digital Supply Chain Twins has been used [1][10][93], we prefer to use “Digital Twins of Supply Chains” (DTofSC). The concept goes beyond the notion of Digital Supply Chain [26][1][93][95][96][97] and envisions full visibility [99] and control across supply chains – e.g., transportation, suppliers, production, sustainability and circularity [1][21] –, closely aligning with so-called control tower models [100][101].

A few real-world examples of DT applied to circular supply chains are gradually emerging in the current literature. For example, a recent study by Valero et al. [102] develops a conceptual framework of DT for a circular meat supply chain, where DT can be applied to help managing and coordinating surplus redistribution between donors and beneficiaries, and the use of food waste for compost production. In another study, Zhihan Lv et al. [103] apply DT concepts to improve the resilience and mitigate losses in the supply chain of protective medical masks. Their model proposes a resilient supply chain’s emergency decision cost model and profit model under the mode of manufacturer sharing supplier. In this paper, we build on circular supply chain visibility and control, considering other supply chain capabilities that can be developed with a DTofSC such as optimisation and simulation [95], resilience and redundancy (e.g., managing ripple effects [98][107]), trust and security, lower costs and efficiency (e.g., inventory management or delivery times [1][26]), and support to circularity and sustainable operations [104]. DTofSCs is also anticipated to offer unparalleled levels of business analytics, which in turn directly maps to risk at different stages of service delivery. Rather close to our vision, we further note a DT approach to services (with a focus on manufacturing but flexible enough to handle other domains) and model-based system engineering [97] with an architecture comprising successive abstraction from physical entities to data fusion. Finally, advanced track and tracing (e.g., with blockchain technology) can bring human-friendly visualisations and also quickly alert for deviations well ahead of disruption [107], particularly if coupled with advances in Artificial Intelligence [15][17].

Three challenges emerge from literature. The first challenge is *how to bind a digital representation to the physical system*, necessarily a combination of IoT and manual intervention [105][15]. To this sense, a number of technologies bridge gaps between the two mirror images – common examples are novel connectivity technologies (5G for wide coverage [84], or Zigbee or Bluetooth for local communications [106]) and embedding product tags (e.g., RFIDs or smart sensors

[20][107]).

The second challenge concerns *reconfigurability* [14] in the sense that DTofSCs must be as adaptive and dynamic as the SCs themselves. Currently, a significant barrier in deploying DTs is that of adapting the DT to changes in the physical settings [14], and the desire is that the DTs technology will detect modifications in configurations and autonomously re-create the DT so to maintain the desired level of fidelity.

A third challenge concerns *integration and interoperability*. To a practical extent, DTofSC should be seen as a generational upgrade of ERPs [18] but now with full real-time response and visualisation and integration with prediction, optimisation and analytics modules.

We argue there is a further fourth challenge, *interoperability and collaboration*: considering that SCs are open, distributed, and multi-party systems, effective means for technical collaboration are essential to realise concept of DTofSCs. For example, a framework needs to exist that will allow multiple logistic operators to discover the technical interfaces so their (typically) different and non-integrated systems can seamlessly collaborate [19][69]. Blockchain technologies can here play an important role [16]. This aspect in itself raises important trust problems around verifiability of information and parties [24]. This is a key topic we will further discuss in Section IV.

C. Data Models and Ontologies

A fundamental element of any DT is the underlying data models: a computer needs a digital representation over which to receive/send data, perform higher-level reasoning (e.g., with Artificial Intelligence), represent and store data, and to render autonomous visualisations that a human can utilise and visually understand [84]. For a simple example, the digital twin of a manufacturing part needs first to be captured in a data model before a Digital Twin of it can be created. Whereas some work exists [94][108], including early attempts at standardisation [112], we note that, generally in the wider area of Supply Chain Management, research is scarce in this crucial direction. Beyond digitally capturing the state and configuration of a physical object or a process, a DTofSC further needs the data models to be flexible since it should allow for quick reconfiguration and promote liveliness [112], i.e. the ability of a system to capture changes and reflect the dynamics of “live” operations, and be self-healing, up to support decision-making with forecasting and simulation.

A particular area that has shown developments, including regulatory drive, is the notion of DPPs [109][111]. A DPP is, in its essence, a data model of a product. A good has information associated with it that is available in a structured format and, hence, both of use for the human user and an automated process. In particular for batteries, the European Union has now mandated [113], that certain batteries (notably Electric Vehicles [114]) must be accompanied by a DPP, in an effort to facilitate circularity [115]. Whereas other attempts are known (e.g., materials [116]) the sustainability and circularity communities seem to be driving the efforts [110], both in terms of efficiency [117][118][119][120] and transparency [120][121][123].

IV. SYNTHESISED SYSTEMS VIEW

In this section we present our second main contribution: a synthesised systems architecture of DTofSCs; while elaborating, we will identify a key gap without which global visibility and actionability of supply chains may remain of limited value – the *addressability requirement*.

A. System Architecture

Beyond *visibility* of Supply Chains, we also envision *actionability* (or control); with these two broad requirements, once can aim at high levels of automation, simulation and forecasting. To this sense, we apply Computer Science methodologies focusing on layered approaches from the physical objects/processes at the bottom to increasingly more abstract (and higher value) services. A detailed diagram of our approach to DTofSCs is shown in Figure 5. It proposes 4 layers, counting from the bottom: the Physical Twin (abbreviated as PhyT), the Convergence Layer (Conv), the Digital Twin (DT), and Higher-Level Applications (HLA).

1) Physical Twin

The PhyT consists of the physical processes associated with the Supply Chain. To note that the supply chain illustrated in Figure 5 is only representative as multiple variations and lengths are possible. We distinguish between two sub-layers. At the bottom, we have the *Supply Chain view*, essentially composed of the parties involved in delivering the product or service, from manufacturers to retailers, depending on the specific supply chain. We note that this sub-layer has important data to offer to the DTofSC as a whole such as provenance, authenticity, etc. Immediately mapped onto the supply chain, we have the *product view* which models the product as it progresses through the supply chain. We annotate each step with, quite generically, the four elements that go into any product: (1) parts and materials, (2) work and labour, (3) capital, and (4) information compliance).

For the parties involved and at each step in the product's journey, we note that data is “pulled” into the higher layers of our model. From the Supply Chain view, we obtain information about the parties; from the Product view we obtain information about the product. The notion of Application Programming Interface (API) [20][94], a common concept in Computer Science, is here essential: there must be means for this data to be pulled into the higher-layers.

A particular requirement is that, for full global visibility and actionability, each party and each step must be uniquely identifiable, similarly to devices on the Internet each with a unique (at any moment) IP address. This notion, that we term the *Addressability Requirement*, has not been fully captured to the best of our knowledge. To a sense, this is the role of a label with a Serial Number or a RFID tag plays. The difference is that these numbering systems are not designed to be global and have only meaning within the organisation that issued them, nor are actionable (e.g., like a Web link). Designing such numbering system is not straightforward and we argue is a critical research direction for full deployment of DTofSCs that the authors are currently tackling.

2) Convergence Layer

The Convergence layer embeds connectivity and discovery of participants and systems. To cope with scale and heterogeneity of systems, we show different (white) clouds representing local systems (likely closed and proprietary) that, using interoperable and open interfaces, connect to an upper global function where all parties meet. This point is where we envision blockchain technologies to be best positioned – it directly reflects Figure 3-top. A blockchain, rather than conveying data by itself, should be seen as a *meeting point* for a single, resilient, trust-less discovery point, which can further enable actionability. It is perfectly possible to deliver these requirements on top of existing ERP systems, however, the coordination effort may be too high to enable full interoperability, as discussed.

Slightly delving into technical implementation, the API endpoints (e.g., addresses) that are offered by the lower levels are here reconciliated. In other words, existing a global infrastructure where such addresses can be found (including using automation and self-discovery), the Convergence layer provides mediation and abstraction so that the lower layers can have any kind of technical implementation as long the interfaces are interoperable. This is common practice in Web-based applications where the functionality is not necessarily public but how the functionality is invoked uses open, interoperable means.

3) Digital Twin

The Digital Twin layer virtualises two aspects. First, it virtualises the Supply Chain in itself with its parties – suppliers, manufacturers, retailers, etc. We here include a supplier discovery function [14], critical to support resilience and manage risk. If all parties offering services in a specific Supply Chain have a virtual presence, and such presence is, technically, supported by industry standards for interoperability, we envision a feature where supplier searching can achieve high levels of automation.

Second, we have the digitalisation of the specific instance of the Supply Chain with two routes: upwards, by querying the current state of the product in its journey (for visibility, e.g., tracing parts), and downwards, actuating on it (e.g., to modify quantity of orders given external events). If one envisions high levels of automation, both functions require, as discussed, interoperable data models; whereas parties need organisational profiles, products need a notion of DPPs.

Security and Trust Considerations— Equally important, and crucial from an enabling perspective for a new approach to Supply Chains, is the security, trust and confidentiality of data and actuation. Given that it may secondarily touch on individual persons personal data, there are further privacy and Data Protection elements to consider. This is generically represented

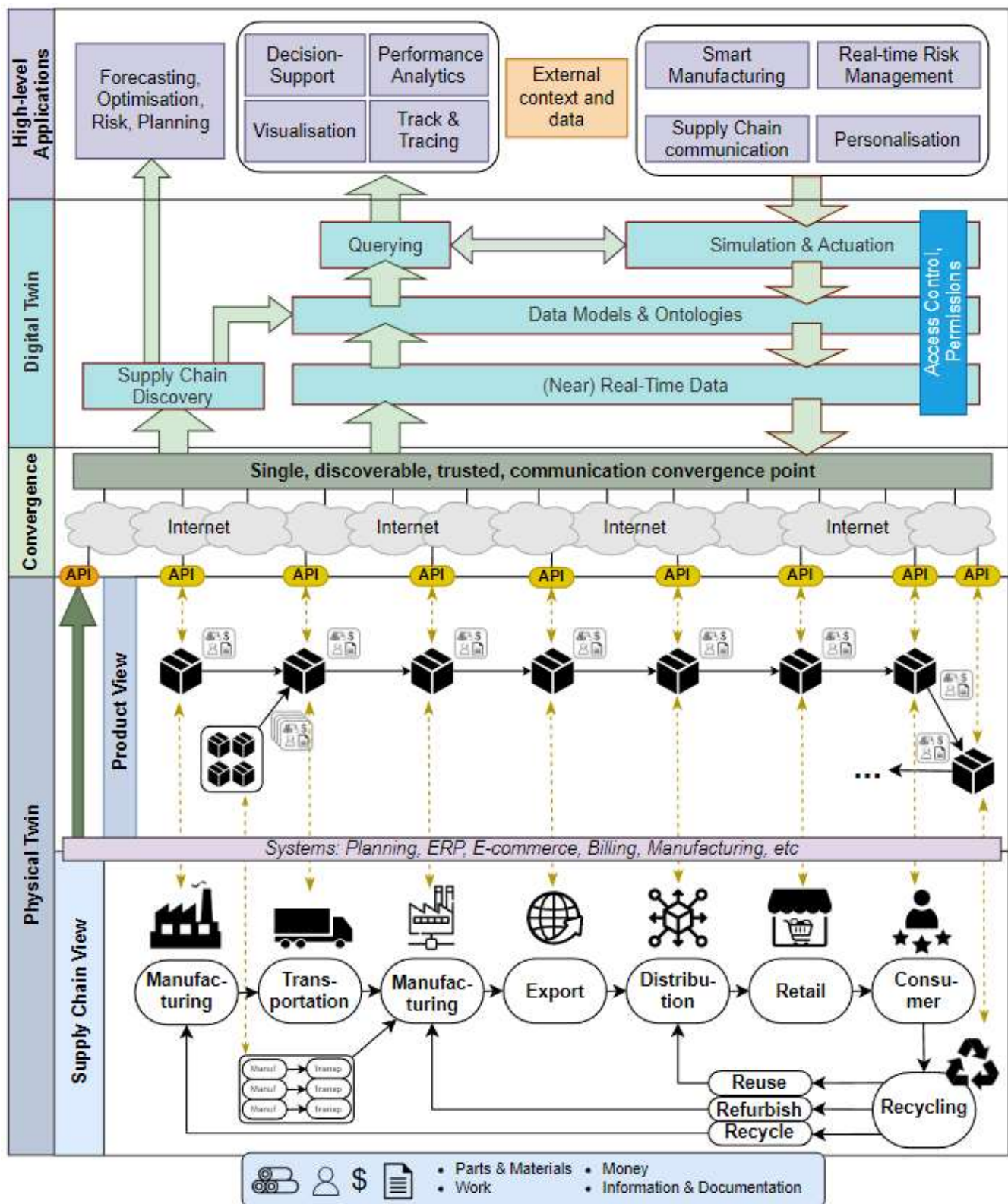


Figure 5. System architecture of Digital Twins of Supply Chains.

by the vertically transversal module named “Access Control, Permissions”. We will not delve deeply in this component given its complexity and necessary technicalities, but, instead, outline key considerations, most to be developed in future (technical)

work.

A first aspect to consider is the deliberate free flow and openness of information: it must come with robust *access control* and permissioning models, which need to be natively

embedded in the framework. The primary motivation is *confidentiality*, but also protecting the organisation against both *accidental disclosure* of internal information and malicious modification (a requirement of *integrity of data* thus necessitating an *identity subsystem*), since our framework contemplates actuation (i.e., not only read-only) on the Supply Chain. In other words, organisations participating in the SC will, necessarily, be mindful of *what* information is accessible and modifiable/appendable to *which* organisations. Establishing a parallel with common models of digital access (e.g., files) we have privileges associated with reading and writing.

Another requirement is *availability*. Beyond robustness against service downtime affecting normal operations, there is the further need of *trusted auditing* such as within a regulated environment. This directly ties to *secure logging and monitoring of changes*, to varying degrees, in the information recorded as an item traverses the Supply Chain.

Related to the previous, and considering the scale and distributedness of the system, there is a need of strict requirements to be enforce on *authenticity* (a requirement involved in Proof of Ownership Management, as discussed in Table 1). This requirement maps to *verifiability and attestation*, where any party (with appropriate access) can verify, and obtain convincing evidence, *using its own means*, to present to third-parties, that the information does originate from a given party. A central case is, upon a party registering an event about an item, downstream parties need means to verify it was registered by a party that has (what we called) the ground truth. For example, whereas a retailer has authority to publish that a product bundle is composed of items received on a certain date, it cannot assert information about the provenance of raw materials. It should, however, be able to verify and attest the identity of the parties who asserted that information.

A related requirement to authenticity, somewhat in reverse, is *non-repudiation*: Whereas authenticity gives means for an honest party to prove they created a set of data, non-repudiation is the requirement that gives means to anybody else to prove a dishonest party did create a set of data despite denying. The technical architecture should perhaps not require this component so organisations not incur in accidental liability; however, this is likely an important requirement for law enforcement and government agencies who may wish to enforce it.

Finally, as generic requirement, the framework should be robust to *cybersecurity attacks*. On the one hand, this falls under generic IT cybersecurity (thus out of scope for this paper); on the other hand, it is interesting to see how Blockchains such as Bitcoin or Ethereum have remarkably resisted cyberattacks, noting that virtually all known did not directly target the blockchain but rather the technical development processes (e.g., vulnerable smart contracts) as discussed in Section III-A.3 – “Human and Managerial Element”.

The consideration of regulatory frameworks to govern the collection, use, and sharing of supply chain data within the context of DTofSC is of critical importance. Once the operational agreements are established, a data governance plan

should address the following [121]:

- Definition of data inputs, processing, and outputs responsibilities.
- Agreement on data privacy and sharing principles.
- Establishment of data authorization, ownership, and accountability norms.
- Agreement of meta-data ownership and liability concerning processed data.

4) Higher-level Applications

We then arrive at the top layer which builds on the rich data this model allows and becomes specific to the use-case. The functions represented in Figure 5 should be seen only as illustrative. Expected features are both reactive and proactive – from forecasting risk and planning inventories, to quick modification of orders based on, e.g., notions such as open supply chains (towards co-production) [15][22], or mass personalisation via Industry 4.0/5.0 [16][17], to embedding circular routes on-demand [14]. It is likely that these higher-level applications are enhanced with external data and/or integrated with internal systems such as ERPs, CRM, E-commerce platforms, Billing systems, planning systems, etc.. Traditional functions can be much augmented (warehouse levels, transport times, vendor performance, route risks, etc.), but new forms of business decision tools are allowed. Risk management, in particular, can take a much more formal and quantitative form. Using risk profiles based on actual analytics and history of performance of vendors, materials, routes, regions, etc., one can perform simulations with modified environmental parameters (e.g., increased rainfall, increased political risk, import sanctions, etc), and embed supplier performance. It therefore presents an opportunity to apply AI techniques given the rich and generous data collected that can be used to train AI models supporting decision processes [23].

B. Application Example: Battery Recycling

As an illustrative example, Figure 6 shows how our architecture (Figure 5) could be applied to recycling of Batteries of Electric Vehicles (EV). The supply chain was modelled after [124][125][126][127] while informed by a workshop the team held with EV and Battery Circularity stakeholders.

1) Overview

As an illustrative example, Figure 6 shows how our architecture could be applied to recycling of Batteries of Electric Vehicles (EV). The supply chain was modelled after [124][125][126][127] while informed by a workshop the authors held with EV and Battery Circularity stakeholders. We kept the system architecture deliberately simple and omitted several (potential) functions and functional elements. At the bottom, we show 7 parties: a Battery Manufacturer supplies an EV manufacturer. This section will also clarify how we view the role of Blockchain Technology. Despite the fact it plays a central role, it is used in a significantly different fashion as per most existing literature.

The battery can be tracked, queried (e.g., retrieve its currently known information), actuated (potentially change its

properties and context), and visualised at any time. If sufficient data is collected at frequent intervals (Figure 6-top), and across the whole supply chain, high-level business analytics become possible that will support higher-level functions such as determining the environmental impact or forecasts and risk assessments. This is accomplished by registering it in a globally accessible system with a unique identifier along with its profile (the DPP). A DPP, when integrated in the EV, should have a (trusted) annotation on its profile to say how it was used and within which EV that should, equally, undergo a similar registration procedure. After selling the vehicle, the owner of the EV will seek maintenance and battery records are updated. When the car reaches the end of its lifecycle, a specialised party will dismantle and extract the battery (potentially following instructions from the original and EV manufacturers. This is represented by the dashed arrows labelled with “profile and attestation”. The used battery will be sent to a party that will test its condition and make a decision about its destination. We represent two routes. One is a battery whose condition is acceptable for repurposing such as energy storage [125] – for example, the battery can still hold at least 50% of the original capacity. If it is deemed that the battery reached end-of-life, its passport is updated and locked, as it will go recycling up to ending as “black mass” [126], raw materials similar to those directly from a mine.

Figure 7 shows a detailed view of the operations when interacting with our model. We use signalling diagrams: vertically we have time, and horizontally we have the parties involved. We take three perspectives:

- the *EV Manufacturer* (“EV Manuf”), imagining a new EV is just being fitted a battery and soon after being sold, who creates a new entity (the battery) and adds its records for others to use
- a *party providing maintenance to an EV* (“EV Maint”), who has special access (given by EV Manufacturer) to data about the battery but who cannot modify (hence being an observer) but can, perhaps, add records of EV maintenance thus complementing the history of the battery (e.g., test results and measured capacity at the time)
- a generic *Public Observer* who is a party that has no special access to the battery information except that originally deemed public by the manufacturer; this may be the case of a generic recycler or a second-hand dealer trying to prove to prospective customers that the battery is the original that came with the vehicle.

2) Local Perspective

We depict three points in the lifecycle of the battery: battery/EV manufacturing, maintenance, and public access.

Manufacturing— Figure 7 assumes a factory where an EV is being assembled and captures the moment when a battery is being fitted. The raw battery has its own DPP which comes with an *address* from the previous party in the SC. Once fit in the new EV, both the new battery product and the EV get new (globally unique, potentially mediated by a blockchain) addresses and DPPs that are linked to all DPPs that the battery, and the EV are composed of. We focus on the battery. The new entity in the Supply Chain is registered in a local system managed by the EV manufacturer. This system needs to be

open, to a degree, to the outside world as it holds the authoritative information about the EV and, in particular, the battery. Such a DPP is annotated with further information, including programmatic methods with which an external system can automatically interact with (e.g., for a specific item, obtain the date it was produced). It is important that this ERP system is managed locally for scalability reasons: no global system would ever cope with the scale of information produced; furthermore, this information is likely to exist already, albeit in formats that are not interoperable and even less generically accessible to third-parties.

The *address* of this information (but not the information itself) is now stored in a globally accessible system – for which a public blockchain seem the perfect fit since an address can be a small piece of data that simply records the creation of a product. This address can be imagined as similar to a URL that can be shared, and the actual information (the content of the website) is now on a server located at that address.

The physical object, the battery, should be physically associated with the address (e.g., if the material allows, an engraving) – this aspect is beyond the scope of this paper and falls into our future work) and we can only conjecture this is generally feasible. The EV is now shipped to a dealer or customer and fully enters the supply chain. This event is recorded in the local repository and can later be retrieved. Finally, access to information about the EV needs to be selectively granted.

Maintenance— We now illustrate how the EV/battery is managed once the product is in use by the end customer. Figure 8 shows the first scenario where there is an authorised party by the original manufacturer to access specific (e.g., technical) information about the battery, along with having the appropriate privileges to add information to the specific battery by adding, e.g., records of maintenance and health. This party has means to retrieve the address of the battery (e.g., its engraved in the casing); using the global, decentralised system (e.g., over a blockchain), the battery profile is looked-up which will take the agent to the original manufacturer’s repository. Since the EV Maintainer has specific access, it will be able to retrieve information and add events to it.

We note there is a variation that may fit better this and other scenarios: the EV maintainer has its own repository of information where the new data about the battery is stored and the blockchain, being the point of convergence, will store a pointer to the new information. It is then up to the application that looks up information to reconcile all the records in a form that users find more helpful.

Generic Public Access— A simpler variation of the former scenarios is shown in Figure 9. It consists of a generic user having access to the address of the battery and looking up information. This scenario could range between a prospective buyer of the (now) second-hand EV (e.g., seller shares information about its vehicle), to a government agency extracting data, at scale, to build statistics about EV and battery usage. The level of access is necessarily limited and shaped by both regulations and the level of openness the involved parties (in this case, the EV Manufacturer and EV Maintainer).

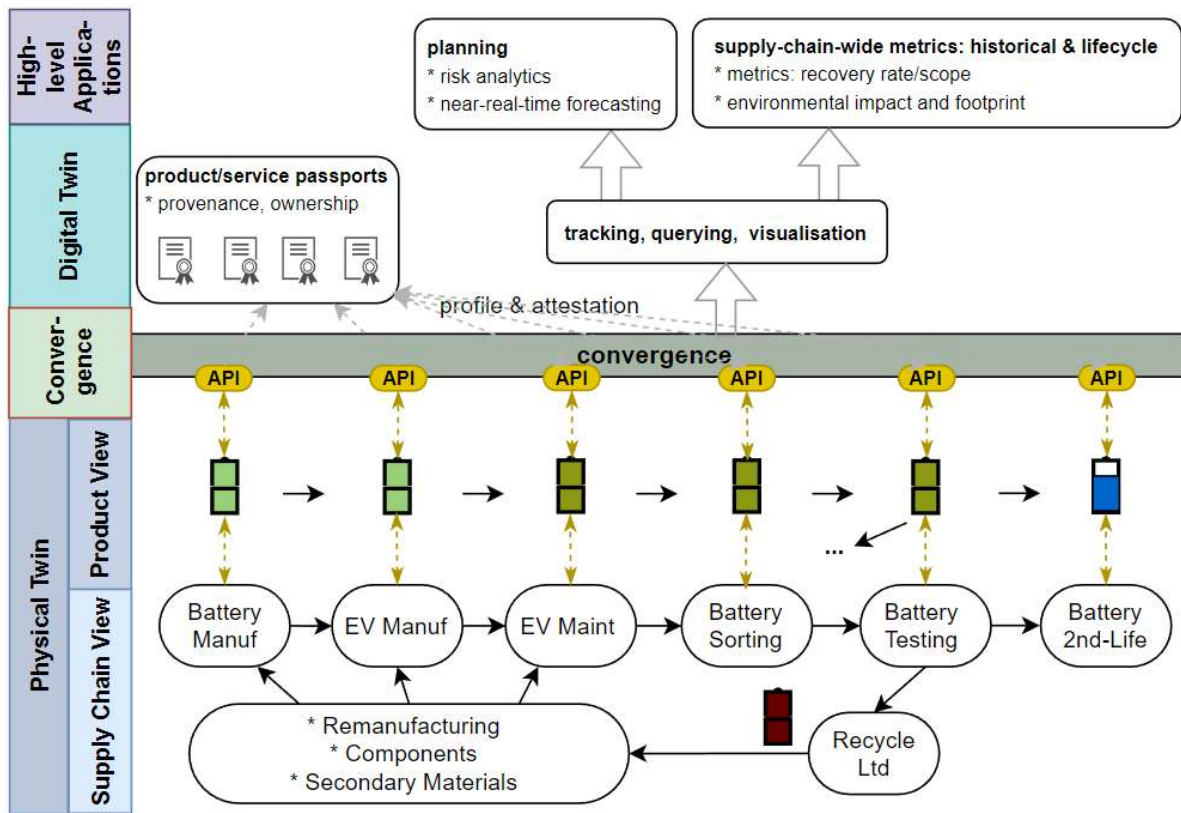


Figure 6. Application of our framework to EV battery recycling.

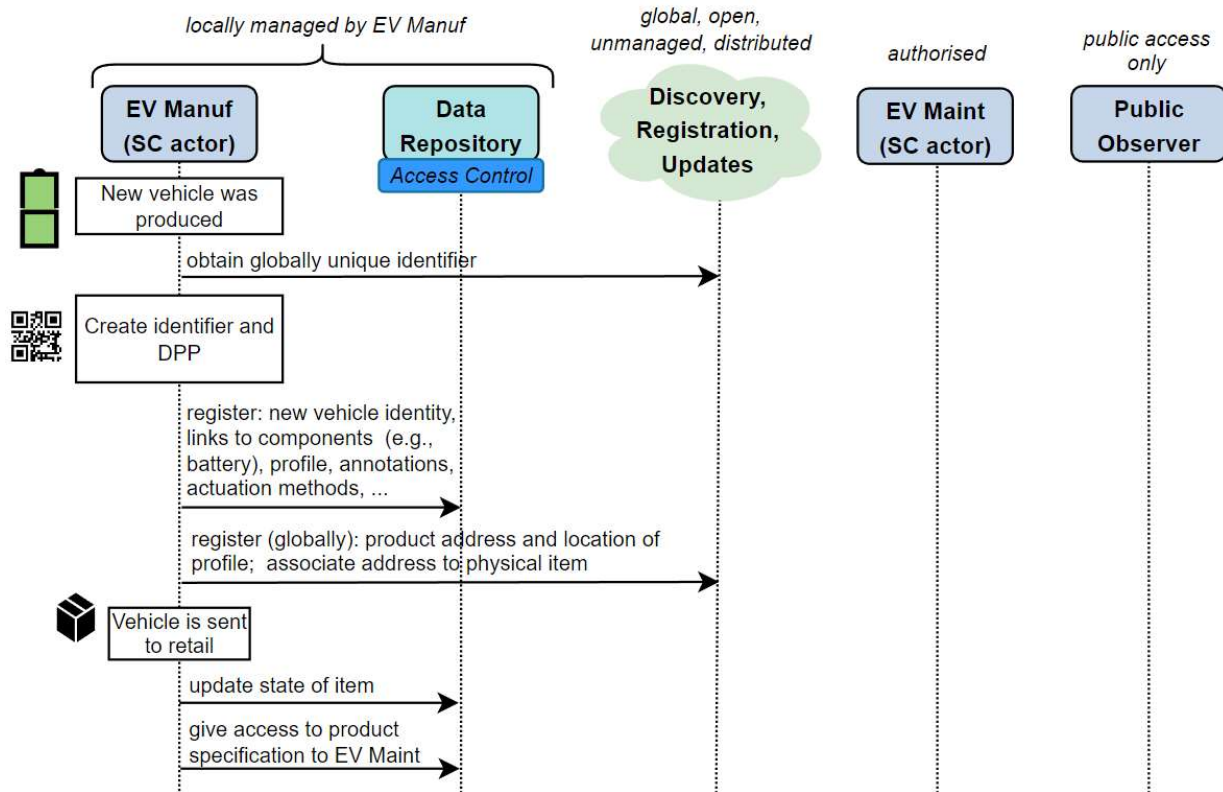


Figure 7. Sequence of operations as per our model and use-case: creation of new entity.

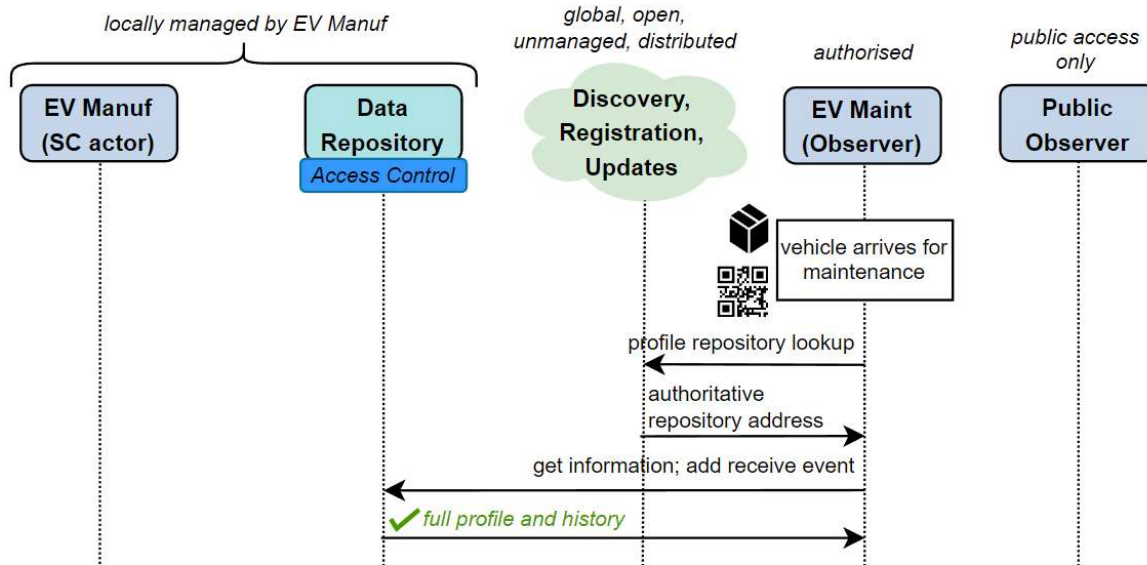


Figure 8. Sequence of operations as per our model and use-case: downstream party obtaining information.

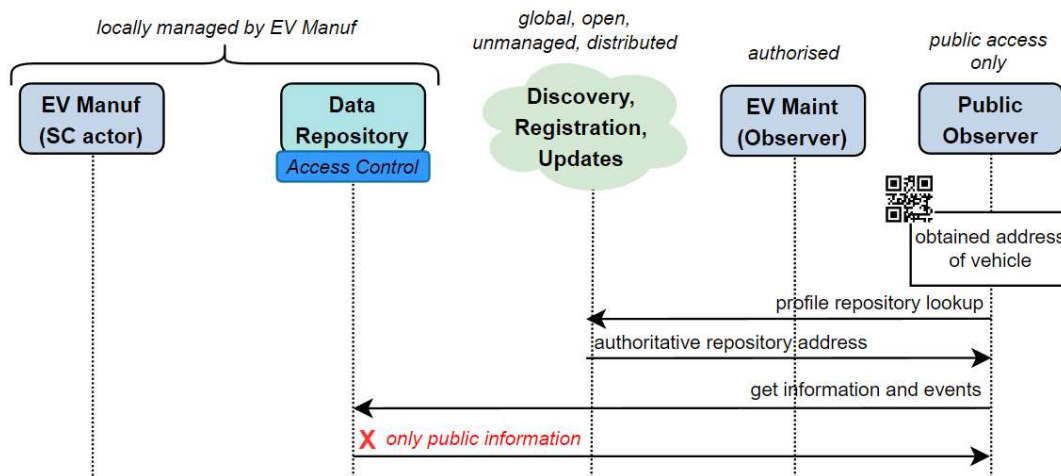


Figure 9. Sequence of operations as per our model and use-case: controlled open access to product information.

We note that the address of the battery is a purely digital object that can be shared, e.g., in a text message.

V. CONCLUSIONS AND OUTLOOK

By reviewing the literature around the topic of Digital Supply Chains, and drawing a novel systems model, we simultaneously attempted a clarification and synthesis of the notion of DTofSC. We have identified the key areas for a systems approach, and applied a computer science design methodology from which we derive a conceptual systems architecture for a globally scalable, accessible and self-coordinated (digital) Supply Chains. On applying it to the case of battery recycling, we show its potential usefulness while shedding light on practical aspects.

We start by stressing that the concept is not strictly novel but, as shown in high-quality literature, it is inconsistent scattered, and only at a basic conceptual level. To further evidence this state of play, there are no real-uses cases for the notion of DTofSCs; rather, there are numerous applications of (classic) Digital Twins that often extend beyond digitalisation of a physical item to encompass, e.g., logistics. At any measure, our and others concept is vastly broader.

Overall, two key findings can be drawn in direct address of our first research question: *To which extent can we generalise the notion of Digital Twins in product design/manufacturing to Supply Chain Management?* Laterally, we address another research objective: *Review and generalise the notion of Digital*

Twins to apply to Supply Chains, aligned with a review of enabling technologies. Our paper offers a comprehensive review of the application of Blockchain Technology to Digital Supply Chains; our conclusions and critique is that the technology is useful, and a game-changer, but needs to be used under a different light and integrated with other technologies in order to satisfy much wider requirements. Above all, applications must always consider (what we call) the EDI model (Figure 3) and only privilege Blockchains if strictly needed.

The second key finding, which is a central contribution of our paper, is that, in order to remove coordination parties, one needs to tackle (what we call) the *addressability requirement*, which will be a focus on our future (technical) work. To the best of our knowledge, we are the first to identify such a requirement.

Whereas a technical elaboration of addressing/identity systems is outside the scope of this paper, we do not anticipate this to be a hard technical challenge, since it ties to the previous consideration (scalability) which addresses our third research objective: *Identify, position and characterise gaps, particularly, what we call the addressability.*

Finally, we note a secondary challenge pertaining to Cybersecurity and Confidentiality, without which any technical design cannot fully meet realistic implementation. Supply chain and operational managers should conduct a comprehensive risk assessment to identify potential privacy and security risks before implement a digital twin system. By doing so they will be able also to develop appropriate measures to mitigate them. Some of these strategies should also include the implementation of a strong authentication and access mechanisms, such as multi-factor authentication and encrypting sensitive data to prevent unauthorized access to the digital twin system. Last but not least there is a need to develop a best practice and educate the employees to ensure they are aware of potential threats and know how to respond to them and to collaborate with various stakeholders such as IT professionals, data privacy experts, and legal advisors, to ensure that privacy and security considerations are addressed effectively throughout the digital twin implementation process.

Outlook— A key research opportunity is creating a simple prototype of a full running system able to track a simple product; for example, a product within the perishable food industry that needs anti-counterfeit measures (e.g., wine). Not only this will allow us to test the integration of a number of different concepts, but we will also be able to develop means to use data analytics and start assessing its social/business potential.

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